FE SEM 2 Rev NEP 2020 Syllabus Scheme for
Elective Chemistry

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- ENGINEERING MATERIALS -

- THEORY SCHEME -

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
BSC2031	Engineering Materials	2		-	2		-	2

		Theory					Term work	Pract / Oral	Total
		Internal Assessment (IAT)			End Sem Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	IAT-I + IAT-II (Total)					
BSC2031	Engineering Materials	15	15	30	45	1.5		--	75

MODULEWISE WEIGHTAGE

Module	Weightage per module
1)Alloys	14
2)Ceramics	13
3)Composites	14
4)Plastics and Elastomers	16
5)Advanced Polymers	11
6)Nano materials	13
Total	**81**

MODULE I : Alloys

(No. of hours – 04, Weightage of marks with option – 14)

- **Course Objective**: To study the composition, properties and functions of various alloys.
- **Course Outcome**: Identify different types of alloys and use them for specific engineering applications.
- **Prerequisite**: Knowledge about purpose of making alloys.

MODULE I : Alloys

Sr. No.	TOPIC	REMARKS
1.	A) Ferrous alloys – Plain-carbon steels, Heat and Shock resisting steels, Stainless steels. Effect of the alloying element- Ni, Cr, Co, Mn, Mo, W and V.	Types and limitations of plain carbon steel General composition, types, properties, and applications of heat, shock resisting steel, stainless steel
2.	B) Aluminium alloys – Composition, properties and uses of i) Duralumin, ii) Magnalium.	---

Sr. No.	TOPIC	REMARKS
3.	Copper alloys – Composition, properties and uses of i) Brass – Dutch Metal and German Silver and ii) Bronze – Gun metal and Nickel bronze.	---
4.	Alloys of Pb – Composition, properties and Uses of i) Wood's metal ii) Tinman's solder.	---
5.	Numerical based on Composition, density, and weight of an alloy	Sample numerical will be provided.

Reference Books (Alloys):

1. Engineering Chemistry, Jain and Jain, Dhanpat Rai Publication.

2. A Textbook of Engineering Chemistry, S. S. Dara, S. Chand and Company.

3. A Textbook of Engineering Chemistry “Engineering Materials & Applications” Shashi Chawla, Dhanpat Rai & Co.

MODULE I : Alloys

- Ferrous Alloys-
- Plain carbon steels:
- Carbon steel is by far the most widely used kind of steel.
- The properties of carbon steel depend primarily on the amount of carbon it contains.

Types of Plain carbon steels

- With an increase in the carbon content from 0.01 to 1.5% in the alloy, its strength and hardness increases but still such an increase beyond 1.5% causes appreciable reduction in the ductility and malleability of the steel.

Low carbon steel or mild steel

- Containing carbon up to 0.25% responds to heat treatment as improvement in the ductility is concerned but has no effect in respect of its strength properties.
- These are possibly produced in the greatest quantities than other alloys. Carbon present in these alloys is limited, and is not enough to strengthen these materials by heat treatment; hence these alloys are strengthened by cold work.

Low carbon steel or mild steel

- Their microstructure consists of ferrite and pearlite, and these alloys are thus relatively soft, ductile combined with high toughness. Hence these materials are easily machinable and weldable.
- Typical applications of these alloys include: structural shapes, tin cans, automobile body components, buildings, etc.

Medium carbon steels

- Having carbon content ranging from 0.25 to 0.70% improves in the machinability by heat treatment. It must also be noted that this steel is especially adaptable for machining or forging and where surface hardness is desirable.
- These are stronger than low carbon steels. However, these are of less ductile than low carbon steels. These alloys can be heat treated to improve their strength.

Medium carbon steels

- Usual heat treatment cycle consists of austenitizing, quenching, and tempering at suitable conditions to acquire required hardness.
- They are often used in tempered condition. As hardenability of these alloys is low, only thin sections can be heat treated using very high quench rates. Ni, Cr and Mo alloying additions improve their hardenability.
- Typical applications include: railway tracks & wheels, gears, other machine parts which may require good combination of strength and toughness.

High carbon steels

- steel-containing carbon in the range of 0.70 to 1.5% and is especially classed as high carbon steel. In the fully heat-treated condition, it is very hard and it will withstand high shear and wear and will thus be subjected to little deformation.
- These are strongest and hardest of carbon steels, and of course their ductility is very limited.
- These are heat treatable, and mostly used in hardened and tempered conditions.

High carbon steels

- They possess very high wear resistance, and capable of holding sharp edges.
- Thus, these are used for tool application such as knives, razors, hacksaw blades, etc.
- With addition of alloying element like Cr, V, Mo, W which forms hard carbides by reacting with carbon present, wear resistance of high carbon steels can be improved considerably.

Limitations of plain carbon steel

- There cannot be strengthening beyond about 100000 psi without significant loss in toughness (impact resistance) and ductility.
- Rapid quench rates are necessary for full hardening in medium-carbon leads to shape distortion and cracking of heat-treated steels.
- Plain-carbon steels have poor impact resistance at low temperatures.
- Plain-carbon steels have poor corrosion resistance for engineering problems.
- Plain-carbon steel oxidises readily at elevated temperatures.

Heat resisting steels

- These alloy steels are suitable making equipments which might be exposed to high temperatures during their usage.
- Important constituents of heat-resisting steels are:

a. Molybdenum: (3.5%), it imparts strength and creep resistance to steel at high temps.

b. Chromium: (12%), increases corrosion resistance at high temperature

c. Nickel: it also improves heat and corrosion resistance.

Shock resisting steels/Spring steels

- It is used where sudden shock-absorption is required
- They possess high elastic limits
- Chrome-Vanadium steels containing 0.8-1.1% chromium, 0.25% vanadium, and 0.35-0.9% carbon possess great impact-resistance
- They are used for making laminated springs, leaf springs, coil springs, etc.

Effect of the alloying element- Ni, Cr, Co, Mn, Mo, W and V

S.N.	Elements	Effects on properties
1	Nickel (Ni)	It improves tensile strength, ductility, toughness, elasticity, heat and corrosion resistances
2	Chromium (Cr)	It improves corrosion resistance, hardness and toughness
3	Cobalt (Co)	It improves hardness, cutting and drilling ability at high speeds and high magnetism.
4	Molybdenum(Mo)	It increases the strength at elevated temperature, corrosion resistance and abrasion resistance.
5	Tungsten(W)	It increases the *red-hardness, toughness, abrasion and shock resistance.
6	Vanadium (V)	Tensile strength, ductility, shock resistance

*The maximum temperature at which steel retains its hardness

Aluminium alloys

Duralumin

Composition: 95% Al, 4% Cu, 0.5% Mg, 0.5% Mn.

Properties:

- Duralumin is a strong, light weighted and hard alloy of aluminum.
- It is also reflective and impermeable.
- It is a malleable metal, and can be easily shaped.
- It is a very good conductor of heat and electricity. It is odorless, and reacts with the oxygen that is around, and forms aluminum oxide.

Aluminium alloys

Duralumin

Properties:

- It is resistant to corrosion.
- They can be easily rolled, folded or forged. They can also be drawn into a variety of shapes and forges.

Uses:

1. It is used for making wire, bar and rods for the screw machine products.
2. It is also used for making die and hand forgings.
3. It is used for making Aircraft structure, truck wheels, screw machine products, rivets and other structural application products.

Aluminium alloys

Magnalium/Magnalumin

Composition: Al: 70-90%, Mg = 30-10%.

Properties:

1. Alloys with smaller amounts of magnesium (about 5%) exhibit greater strength, greater corrosion resistance, and lower density than pure aluminium. Such alloys are also more workable and easier to weld than pure aluminium.
2. Alloys with high amounts of magnesium (around 50%) are brittle and more susceptible to corrosion than aluminium.

Aluminium alloys

Magnalium/Magnalumin

Uses:

1. It is used for making cheap balances, airplane parts, and scientific instruments.
2. It is used in pyrotechnics as a metal fuel and to produce sparks.

Copper alloys

Brass – Dutch Metal

Composition: Cu: 80%, Zn = 20%.

Properties:

1. It has golden colour.
2. It is suitable for all drawing and forming operations.

Uses:

1. Cheap jewellery, musical instruments, battery caps, flexible hoses, tubes, nameplates, etc.

Copper alloys

Brass – German Silver

Composition: Cu: 25-50%, Zn = 10-35%, Sn = 5-35%.

Properties:

1. It possesses good strength and corrosion resistance to salt water.
2. It is extremely ductile, malleable, and looks like silver.

Uses:

1. utensils, tablewares, bolts, screws, ornaments, cutlery, corrosion-resistant, decorative articles, etc.

Copper alloys

Bronze – Gun metal

Composition: Cu: 85%, Zn = 4%, Sn = 8%, Pb=3%.

Properties:

1. It is hard, tough, strong to resist the force of explosion.

Uses:

1. Foundry works, hydraulic fittings, heavy-load bearings, parts of high-pressure steam plants, water fittings, etc.

Copper alloys

Bronze –Nickel bronze

Composition: Cu: 90%, Ni = 9%, Fe = 1%.

Properties:

1. They are hard, higher in tensile strength, better corrosion resistant than copper.

Uses:

1. For rolling purposes, unhardened shafts, valves, and general purpose semi-hard bearings.

Alloys of Pb

Wood's metal

Composition: Bi: 50%, Pb = 25%, Sn = 12.5%, Cd= 12.5%.

Properties:

1. It is readily fusible.

Uses:

1. For making fire alarms and automatic sprinklers
2. For making safety plugs for cookers, milk pot, etc.
3. As castings for dental works
4. For electric and electric fuses, etc.

Alloys of Pb

Tinmann‘s solder

Composition: Sn = 66%, Pb = 34%.

Properties:

1. It has a melting point of around 217°C.
2. Tinmann solder is ideal for applications that have delicate temperature requirements.

Uses:

1. Tinmann solder is commonly used in electronic assembly.
2. It is used for soldering and tinning.

Numerical based on Alloys

Problem 1: Calculation of Alloy Composition

Problem:

An alloy is composed of 70% copper (Cu) and 30% zinc (Zn) by weight. Calculate the atomic percentage of each element in the alloy.

Solution:

1. **Convert percentages to mass fractions:**
   - Mass fraction of Cu (w_{Cu}) = 70% = 0.70
   - Mass fraction of Zn (w_{Zn}) = 30% = 0.30
2. **Calculate atomic percentages:**
   - Assume the atomic masses (molar masses):
     - Atomic mass of Cu (M_{Cu}) ≈ 63.55 g/mol
     - Atomic mass of Zn (M_{Zn}) ≈ 65.38 g/mol
   - Calculate the molar masses of Cu and Zn in the alloy:

$$M_{\mathrm{alloy}} = w_{\mathrm{Cu}} \times M_{\mathrm{Cu}} + w_{\mathrm{Zn}} \times M_{\mathrm{Zn}}$$

$$M_{\text{alloy}} = 0.70 \times 63.55\,\text{g/mol} + 0.30 \times 65.38\,\text{g/mol}$$
$$M_{\text{alloy}} = 44.485 + 19.614$$
$$M_{\text{alloy}} = 64.099\,\text{g/mol}$$

- Calculate the atomic percentages (x_{Cu} and x_{Zn}):

$$x_{\text{Cu}} = \frac{w_{\text{Cu}} \times M_{\text{Cu}}}{M_{\text{alloy}}} \times 100\%$$
$$x_{\text{Cu}} = \frac{0.70 \times 63.55\,\text{g/mol}}{64.099\,\text{g/mol}} \times 100\%$$
$$x_{\text{Cu}} = \frac{44.485}{64.099} \times 100\%$$
$$x_{\text{Cu}} \approx 69.43\%$$

$$x_{\text{Zn}} = \frac{w_{\text{Zn}} \times M_{\text{Zn}}}{M_{\text{alloy}}} \times 100\%$$
$$x_{\text{Zn}} = \frac{0.30 \times 65.38\,\text{g/mol}}{64.099\,\text{g/mol}} \times 100\%$$
$$x_{\text{Zn}} = \frac{19.614}{64.099} \times 100\%$$
$$x_{\text{Zn}} \approx 30.57\%$$

Therefore, the atomic percentages of Cu and Zn in the alloy are approximately 69.43% and 30.57%, respectively.

Problem 2: Calculation of Density of Alloy

Problem:

An alloy is composed of 80% nickel (Ni) and 20% copper (Cu) by weight. Calculate the density of the alloy given that the density of Ni is $8.9\,\mathrm{g/cm^3}$ and the density of Cu is $8.96\,\mathrm{g/cm^3}$.

Solution:

1. **Convert percentages to mass fractions:**
   - Mass fraction of Ni (w_{Ni}) = 80% = 0.80
   - Mass fraction of Cu (w_{Cu}) = 20% = 0.20
2. **Calculate the density of the alloy:**
   - Assume the volumes occupied by Ni and Cu in the alloy are proportional to their mass fractions.
   - Density of the alloy (ρ_{alloy}):

$$\rho_{\mathrm{alloy}} = w_{\mathrm{Ni}} \times \rho_{\mathrm{Ni}} + w_{\mathrm{Cu}} \times \rho_{\mathrm{Cu}}$$
$$\rho_{\mathrm{alloy}} = 0.80 \times 8.9\,\mathrm{g/cm^3} + 0.20 \times 8.96\,\mathrm{g/cm^3}$$
$$\rho_{\mathrm{alloy}} = 7.12\,\mathrm{g/cm^3} + 1.792\,\mathrm{g/cr}$$
$$\rho_{\mathrm{alloy}} = 8.912\,\mathrm{g/cm^3}$$

Problem 3: Calculation of Weight of Components in an Alloy

Problem:

An alloy sample weighs 500 g and consists of 60% iron (Fe), 30% chromium (Cr), and 10% nickel (Ni) by weight. Calculate the weights of each component in the alloy.

Solution:

1. **Convert percentages to mass fractions:**

   - Mass fraction of Fe (w_{Fe}) = 60% = 0.60
   - Mass fraction of Cr (w_{Cr}) = 30% = 0.30
   - Mass fraction of Ni (w_{Ni}) = 10% = 0.10

2. **Calculate the weights of each component:**

   - Given total mass of the alloy = 500 g.

   - Weight of Fe:
     $$\text{Weight of Fe} = w_{\text{Fe}} \times \text{Total mass of alloy}$$
     $$\text{Weight of Fe} = 0.60 \times 500\,\text{g}$$
     $$\text{Weight of Fe} = 300\,\text{g}$$

   - Weight of Cr:
     $$\text{Weight of Cr} = w_{\text{Cr}} \times \text{Total mass of alloy}$$
     $$\text{Weight of Cr} = 0.30 \times 500\,\text{g}$$
     $$\text{Weight of Cr} = 150\,\text{g}$$

   - Weight of Ni:
     $$\text{Weight of Ni} = w_{\text{Ni}} \times \text{Total mass of alloy}$$
     $$\text{Weight of Ni} = 0.10 \times 500\,\text{g}$$
     $$\text{Weight of Ni} = 50\,\text{g}$$

Therefore, the weights of Fe, Cr, and Ni in the alloy are 300 g, 150 g, and 50 g, respectively.

These problems and solutions demonstrate va[illegible]s calculations related to alloys, including composition, density, and component weights.